

# Morphological Analysis for Indian Languages

Technical Presentation | Assessment Tool 2 | Question 12

How word structure, language-specific rules and NLP techniques help process Tamil, Hindi and Telugu.

தமிழ் Tamil

हिन्दी Hindi

తెలుగు Telugu

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*One word  $\rightarrow$  root + affix + grammar*

# What is Morphological Analysis?

Breaking a word into meaningful units to understand its structure.



## Morphology

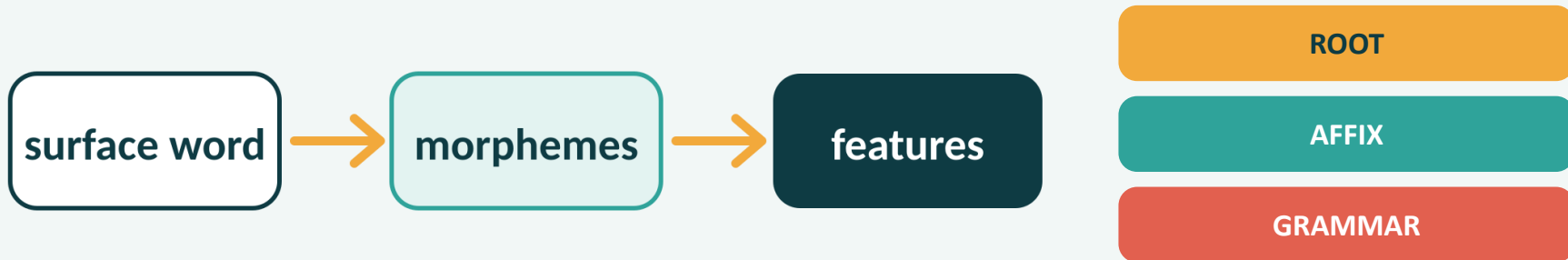
The study of how words are formed and how smaller meaningful units — morphemes — combine.



## Why NLP needs it

A word can carry information about meaning, tense, number, case or other grammatical features.

### EXAMPLE IDEA



# Why Indian Languages Need Special Attention

Question 12 highlights morphologically rich languages such as Tamil, Hindi and Telugu.

## INDIAN LANGUAGE DIVERSITY



தமிழ்

Tamil



हिन्दी

Hindi



తెలుగు

Telugu



### Rich word formation

A word may encode several pieces of linguistic information.



### Many surface forms

The same underlying idea can appear in multiple forms.



### Language-specific rules

Morphological patterns vary between languages.



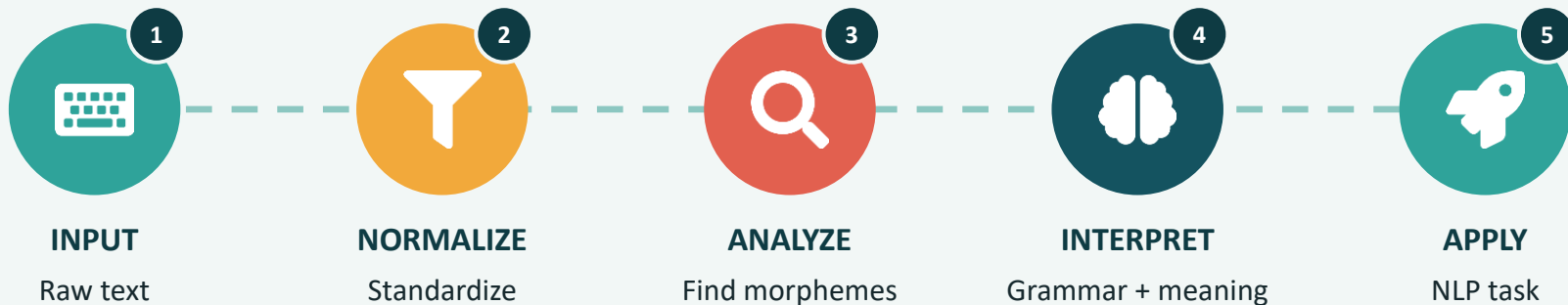
### Limited resources

Rules, dictionaries and annotated datasets may be needed.

**Key idea:** NLP cannot treat every Indian language as if it follows the same word-formation rules.

# A Practical Morphological Processing Pipeline

A simple language-aware flow from raw text to useful NLP features.



## Rule-based

Explicit linguistic rules are useful when patterns are well understood.



## Finite-state

Finite-state models can represent structured word-level transformations.



## Data-driven

Annotated resources can help models learn recurring morphological patterns.

# Where Morphological Analysis Helps

The output can become useful information for several NLP tasks.



## SEARCH

Connect related word forms.



## CLASSIFY

Create better linguistic features.



## TRANSLATE

Handle grammatical differences.



## LANGUAGE TOOLS

Support normalization and analysis.



# What Should We Remember?

A concise conclusion for the class presentation.



01

## Structure matters

Morphological analysis reveals information hidden inside word forms.



02

## Languages differ

Tamil, Hindi and Telugu require language-aware processing approaches.



03

## Better NLP

Good morphology improves normalization, recognition and downstream tasks.

RAW WORD

MORPHEMES

LINGUISTIC FEATURES

NLP APPLICATION

*Morphological analysis connects the surface form of a word with the linguistic information an NLP system needs.*

THANK YOU

# Questions?

தமிழ்

हिन्दी

తెలుగు

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